Date: 2080/12/21
Class: BSc 2nd year
Roll No: 68
Shift: Day

Physics Practical Sheet

Prithvi Narayan Campus

Experiment No: 6 (B)
Group: P1
Sub: Physics
Set: 1

Object of the Experiment (Block Letters): TO VERIFY NOR GATE AS AN UNIVERSAL GATE USING IC-7402.

APPARATUS REQUIRED:

(i) Diode (ii) Bread board (iii) Wires
(iv) Multimeter (v) IC (7402) (vi) Transistor

THEORY:

The NOR gates can be used to perform the basics logic functions of OR, AND and NOT gates and NAND gates also. This is why it's often known as universal gates.

(I) OR gate:

As shown in the figure, the output from the NOR gate is $\overline{A+B}$. By using another NOT gate in the output, the final output is inverted and the final result is given by,

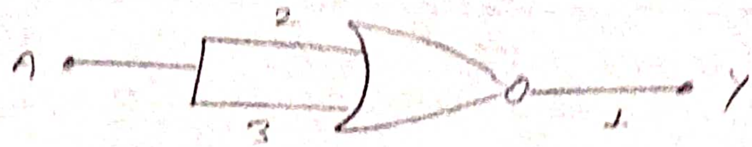
$$Y = A+B$$

(II) AND gate:

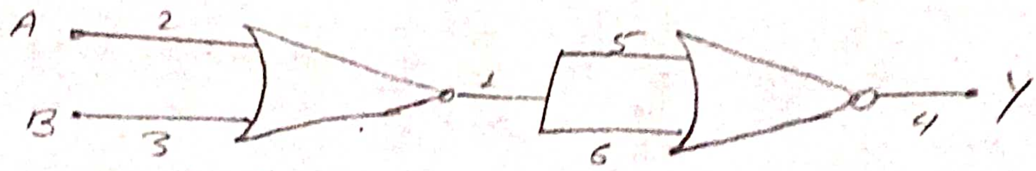
As shown in the figure, the input A and B are inverted by two NOT gates before they are applied to the NOR gate. The output is $\overline{\bar{A}+\bar{B}}$ which is equal to $Y = A \cdot B$ by de-Morgan's Theorem.

(III) NOT gate:

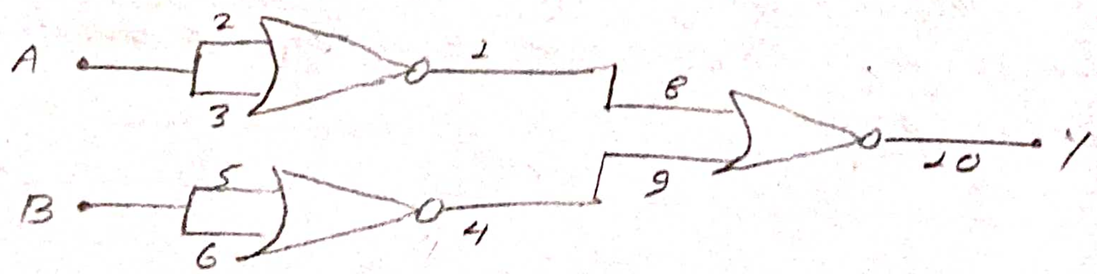
As shown in the figure, the output $\bar{A}$ can be obtained when a single input A is used.



(I) NOR as NOT gate



(II) NOR as OR gate



(III) NOR as AND gate

Fig: NOR gate as Universal gate

OBSERVATION:

(I) OR gates:

S.N	Inputs		Output
	A	B	Y = A+B
1.	0	0	0
2.	0	4.47	4.42
3.	4.47	0	4.45
4.	4.47	4.47	4.41

(II) AND gate:

S.N	Inputs		Output
	A	B	$Y = A \cdot B$
1.	0	0	0.02
2.	0	3.80	0.02
3.	3.80	0	1.88
4.	3.80	3.80	5.09

(III) NOT gate:

S.N	Input	Output
	A	$Y = \bar{A}$
1.	0	4.45
2.	4.34	0

RESULT:

Hence, from the above observation table it is verified that NOR gate is an universal gate.

PRECAUTIONS:

(i) Loose connection should be avoided.

(ii) Power supply of 3V should be used.

(iii) Voltage should be noted correctly.